DSI LCD MODULE

SPECIFICATION

Model:	UEDX10260070-DSI-A
Version:	V1.0
Date:	2026.01.14

Customer Confirmation

Approved by	Notes

Please return one of the copies of the specification with your signature to us within two weeks after you receive this document. If it is not returned, we will assume that you agree to the entire contents of this specification document.

VIEWE Confirmation

Prepared by	Reviewed by	Approved by

REVISION HISTORY

Revision	Date	Contents of Revision Change	Remarks
V1.0	2026.01.14	Preliminary release	All

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1. Product Overview

Picture

This product is an integrated 7-inch DSI display module, combining the UE070WS-RB30-A106B LCD screen with a dedicated DSI adapter board. It features dual interface options (15-pin and 22-pin DSI) for compatibility with a wide range of development boards including Raspberry Pi 4B/5B and ESP32-P4, offering plug-and-play connectivity. The module includes an onboard 5V-to-3.3V power circuit and can be powered either through the power terminal or directly via the interfaces.

2. Key Features

2.1 Display Features

1) Screen Size: 7.0-inch IPS panel
2) Resolution: 1024 × 600
3) Interface Mode： MIPI
4) Driver IC： JD9165BA ；
5) Touch IC DRIVER: GT911 (G+G)；
6) More information：

Item	Specification	Unit	Remark
Pixel Driving element	TFT	-	-
Screen Size	7.0	Inch	Diagonal
Resolution	1024(RGB)X600	Dots	-
Active Area	154.21(H)X85.92 (V) mm	mm	-
Pixel pitch (W*H)	0.0502(H) X RGB X 0.1432(V) mm	mm	-
Module Size (W*H*D)	164.9 (H)X100.0(V)X3.5(T)mm	mm	-
Interface Mode	15Pin /22Pin MIPI	-	-
Luminance	350	cd/m^2	Typ.
Viewing Direction	ALL	O'clock	-
Display Color	16.7M	Colors	24bits
Light Source	LED Backlight	-	-
NTSC	50%	-	-
Pixel Per inch	169	PPI	

2.2 Interface Features

1) Dual Interface Options:
   a. 15-pin MIPI-DSI (2 data lanes) for ESP32-P4 and Raspberry Pi 4B
   b. 22-pin MIPI-DSI (4 data lanes) for Raspberry Pi 5B
2) Touch Interface: Integrated capacitive touch screen (I2C, GT911 controller)
3) Power Supply Options:
   a. 5V DC input (via power terminal J1)
   b. 3.3V DC input (via interface)
4) Connector Types: FPC connectors, 0.5mm pitch

2.3 Power Characteristics

5) Input Voltage: 5V ±10% (via power terminal)
6) Output Voltage: 3.3V (internal conversion)
7) Conversion Efficiency: >90% (Typical)

2.4 Environmental Features

1) Operating Temperature: -20°C ~ 70°C
2) Storage Temperature: -30°C ~ 80°C
3) Operating Humidity: 10% ~ 90% RH (non-condensing)
4) LED Lifetime: 30,000 hours (to 50% brightness)

2.5 Compatibility

1) 15-pin Interface: ESP32-P4, Raspberry Pi 4B (native compatibility)
2) 22-pin Interface: Raspberry Pi 5B (native compatibility)
3) Other embedded platforms supporting MIPI-DSI interface

2.6 Compatible Product Display

3. Interface Definitions

3.1 15-pin DSI Interface (J2) - for ESP32-P4 and Raspberry Pi 4B

PIN	Signal Name	Type	Level	Description
1	3V3	Power	3.3V	3.3V Power Input/Output, becomes output when 5V terminal is used
2				
3	GND	Power	0V	Ground
5	TP_SDA	Bidirectional	3.3V LVCMOS	Touch panel I2C data, with 4.7kΩ pull-up to 3.3V
5	TP_SCL	Bidirectional	3.3V LVCMOS	Touch panel I2C clock, with 4.7kΩ pull-up to 3.3V
6	GND	Power	0V	Ground
7	MIPI_D0P	Input	MIPI	MIPI data lane 0 (positive)
8	MIPI_D0N	Input	MIPI	MIPI data lane 0 (negative)
9	GND	Power	0V	Ground
10	MIPI_CLKP	Input	MIPI	MIPI clock (positive)
11	MIPI_CLKN	Input	MIPI	MIPI clock (negative)
12	GND	Power	0V	Ground
13	MIPI_D1P	Input	MIPI	MIPI data lane 1 (positive)
14	MIPI_D1N	Input	MIPI	MIPI data lane 1 (negative)
15	GND	Power	0V	Ground

3.2 22-pin DSI Interface (J3) - for Raspberry Pi 5B

PIN	Signal Name	Type	Level	Description
1	GND	Power	0V	Ground
2	MIPI_D0N	Input	MIPI	MIPI data lane 0 (negative)
3	MIPI_D0P	Input	MIPI	MIPI data lane 0 (positive)
4	GND	Power	0V	Ground
5	MIPI_D1N	Input	MIPI	MIPI data lane 1 (negative)
6	MIPI_D1P	Input	MIPI	MIPI data lane 1 (positive)
7	GND	Power	0V	Ground
8	MIPI_CLKN	Input	MIPI	MIPI clock (negative)
9	MIPI_CLKP	Input	MIPI	MIPI clock (positive)
10	GND	Power	0V	Ground
11	MIPI_D2N	Input	MIPI	MIPI data lane 2 (negative)
12	MIPI_D2P	Input	MIPI	MIPI data lane 2 (positive)
13	GND	Power	0V	Ground
14	MIPI_D3N	Input	MIPI	MIPI data lane 3 (negative)
15	MIPI_D3P	Input	MIPI	MIPI data lane 3 (positive)
16	GND	Power	0V	Ground
17	NC	-	-	Not connected
18	NC	-	-	Not connected

PIN	Signal Name	Type	Level	Description
19	GND	Power	0V	Ground
20	TP_SCL	Bidirectional	3.3V LVCMOS	Touch panel I2C clock
21	TP_SDA	Bidirectional	3.3V LVCMOS	Touch panel I2C data
22	3V3	Power	3.3V	3.3V Power Input/Output, becomes output when 5V terminal is used

3.3 5V Power Terminal (J1)

PIN	Signal Name	Description
1	VBAT-5V	5V Power Input (4.5V ~ 5.5V) - **Required for backlight**
2	GND	Power ground

3.4 Important Power Supply Considerations

Critical Information:

1. Backlight Requires 5V: The backlight LED array requires approximately 9V (3 LEDs in series × 3V each), which is generated from the 5V input using a boost converter circuit.

2. 3.3V Only Powers Logic: When using only 3.3V interface power, the display logic will function, but the backlight will not illuminate.

3. Recommended Configuration: Always use the 5V power terminal (J1) for full functionality including backlight.

4. Electrical Characteristics

4.1 Absolute Maximum Ratings

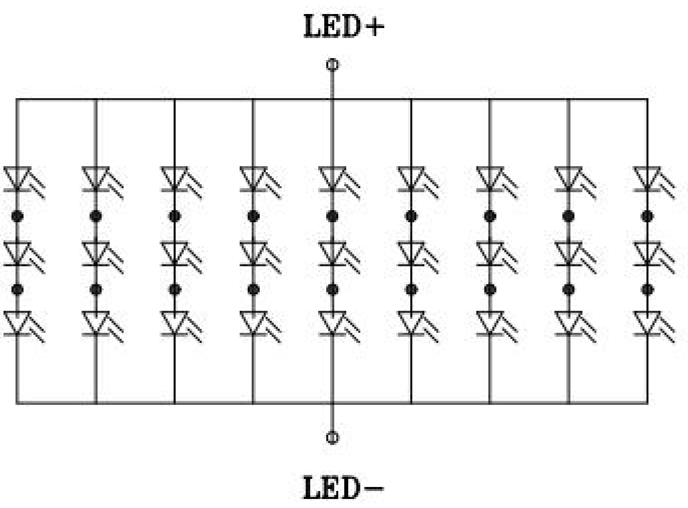
Item	Symbol	Min.	Max.	Unit	Remark
Power supply voltage	VCC	-0.3	4.6	V	Note1
LED forward current	I_F	-0.001	30	mA	For each led,Note1
LED Reverse Voltage	V_R	-	9	V	For each led,Note1
Operating temperature	T_{op}	-20	70	°C	Note1,2
Storage temperature	T_{st}	-30	80	°C	Note1,2
Humidity	H_{st}	10	90	%RH	Note1,3

(Ta=+25°C,GND=0V)

Note1:If the module exceeds the absolute maximum ratings, it may be damaged permanently. Also if the module operates with the absolute maximum ratings for a long time, the reliability may drop.

Note2: In case of temperature below 0℃,the response time of liquid crystal (LC) becomes slower and the color of panel darker than normal one.

Note3: Temp. ≤ 60℃ , 90% RH MAX.
Temp. ＞60℃ , Absolute humidity shall be less than 90% RH.



3串9并，3*3.3=9.9V(MAX),20*9=180mA

4.4 Recommended Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit	Remark
5V Input Voltage	VBAT	4.5	5.0	5.5	V	**Required for backlight**
3.3V Input Voltage	VCC	3.0	3.3	3.6	V	Logic power only
Operating Temperature	Ta	-20	+25	+70	°C	

4.5 Power Consumption Characteristics

5V Power Mode (Full functionality):

Parameter	Condition	Typ.	Max.	Unit	Remark
5V Input Current	Full brightness, white screen	550	600	mA	**Key parameter**
5V Standby Current	Backlight off	140	180	mA	Backlight not connected
Conversion Efficiency	Full load	>90	-	%	LP3321B6F
5V Input Power	Full brightness	2.75	3	W	Estimated

Power Calculation Notes:

1. **5V Power Efficiency**: LP3321B6F conversion efficiency >90%, 5V input power ≈ total power / 0.9
2. **Power Requirements:**
   (1) Voltage: 5V ±5% (4.75V ~ 5.25V)
   (2) Continuous Current Capacity: Recommended ≥1.0A to meet peak demand and leave a margin.
   (3) Recommendation: Use a power adapter capable of providing 5V/1.5A (7.5W) or higher specifications to ensure system stability and long-term reliability.

5. LCD Screen Initialization Parameters

5.1 Screen Timing Parameters

```
dpi_clock_freq_mhz = 50,        // DPI clock frequency: 50MHz
h_size = 1024,                  // Horizontal resolution: 1024 pixels
v_size = 600,                   // Vertical resolution: 600 pixels
hsync_back_porch = 136,         // Horizontal back porch: 136 pixels
hsync_pulse_width = 20,         // Horizontal sync pulse width: 20 pixels
hsync_front_porch = 160,        // Horizontal front porch: 160 pixels
vsync_back_porch = 12,          // Vertical back porch: 12 lines
vsync_pulse_width = 2,          // Vertical sync pulse width: 2 lines
vsync_front_porch = 20,         // Vertical front porch: 20 lines
```

5.2 Initialization Command Sequence

Complete initialization command sequence for JD9165 chip, optimized for ESP32P4 platform (supports only MIPI 2 lanes):

```
// Command format: {command code, {data}, data length, delay_ms}
{0x30, {0x00}, 1, 0},                       // Switch to page 0
{0xF7, {0x49, 0x61, 0x02, 0x00}, 4, 0},     // Chip ID/version settings

{0x30, {0x01}, 1, 0},                       // Switch to page 1
{0x04, {0x0C}, 1, 0},                       // Power control 1
{0x05, {0x08}, 1, 0},                       // Power control 2

// MIPI DSI configuration - ESP32 specific (2 lanes only)
{0x0B, {0x11}, 1, 0},                       // Set MIPI to 2 lanes mode (0x11=2lanes,
0x12=3lanes, 0x13=4lanes)
{0x20, {0x04}, 1, 0},                       // Lane selection register
{0x1F, {0x00}, 1, 0},                       // MIPI HS settle time (0x05->0x00)

// Power and bias settings
{0x23, {0x38}, 1, 0},                       // VCOM control
{0x28, {0x18}, 1, 0},                       // Source driver pre-charge control
{0x29, {0x29}, 1, 0},                       // Source driver main charge control
{0x2A, {0x01}, 1, 0},                       // Source driver main charge control
{0x2B, {0x29}, 1, 0},                       // Source driver main charge control
{0x2C, {0x01}, 1, 0},                       // Source driver main charge control

{0x30, {0x02}, 1, 0},                       // Switch to page 2 (Gamma settings)
// Gamma correction parameters (11 sets of RGB gamma values)
{0x00, {0x05}, 1, 0},
```

```
{0x01, {0x22}, 1, 0},
{0x02, {0x08}, 1, 0},
{0x03, {0x12}, 1, 0},
{0x04, {0x16}, 1, 0},
{0x05, {0x64}, 1, 0},
{0x06, {0x00}, 1, 0},
{0x07, {0x00}, 1, 0},
{0x08, {0x78}, 1, 0},
{0x09, {0x00}, 1, 0},
{0x0A, {0x04}, 1, 0},

// Gamma table data (11 bytes per group)
{0x0B, {0x16,0x17,0x0B,0x0D,0x0D,0x0D,0x11,0x10,0x07,0x07,0x09}, 11, 0},
{0x0C, {0x09,0x1E,0x1E,0x1C,0x1C,0x0D,0x0D,0x0D,0x0D,0x0D,0x0D}, 11, 0},
{0x0D, {0x0A,0x05,0x0B,0x0D,0x0D,0x0D,0x11,0x10,0x06,0x06,0x08}, 11, 0},
{0x0E, {0x08,0x1F,0x1F,0x1D,0x1D,0x0D,0x0D,0x0D,0x0D,0x0D,0x0D}, 11, 0},
{0x0F, {0x0A,0x05,0x0D,0x0B,0x0D,0x0D,0x11,0x10,0x1D,0x1D,0x1F}, 11, 0},
{0x10, {0x1F,0x08,0x08,0x06,0x06,0x0D,0x0D,0x0D,0x0D,0x0D,0x0D}, 11, 0},
{0x11, {0x16,0x17,0x0D,0x0B,0x0D,0x0D,0x11,0x10,0x1C,0x1C,0x1E}, 11, 0},
{0x12, {0x1E,0x09,0x09,0x07,0x07,0x0D,0x0D,0x0D,0x0D,0x0D,0x0D}, 11, 0},

// Display configuration
{0x13, {0x00,0x00,0x00,0x00}, 4, 0},
{0x14, {0x00,0x00,0x41,0x41}, 4, 0},
{0x15, {0x00,0x00,0x00,0x00}, 4, 0},
{0x17, {0x00}, 1, 0},
{0x18, {0x85}, 1, 0},
{0x19, {0x06,0x09}, 2, 0},
{0x1A, {0x05,0x08}, 2, 0},
{0x1B, {0x0A,0x04}, 2, 0},
{0x26, {0x00}, 1, 0},
{0x27, {0x00}, 1, 0},

{0x30, {0x06}, 1, 0},                    // Switch to page 6
// Additional gamma adjustment parameters (14 bytes)
{0x12, {0x3F,0x26,0x27,0x35,0x2D,0x34,0x3F,0x3F,0x3F,0x35,0x2A,0x20,0x16,0x08}, 14, 0},
{0x13, {0x3F,0x26,0x28,0x35,0x27,0x29,0x29,0x2F,0x35,0x2F,0x26,0x20,0x16,0x08}, 14, 0},

{0x30, {0x0A}, 1, 0},                    // Switch to page 10
{0x02, {0x4F}, 1, 0},
{0x0B, {0x40}, 1, 0},

{0x30, {0x0D}, 1, 0},                    // Switch to page 13
```

```
{0x0D, {0x04}, 1, 0},                 // MIPI related settings (0x0C, 0x04)
{0x10, {0x0C}, 1, 0},
{0x11, {0x0C}, 1, 0},
{0x12, {0x0C}, 1, 0},
{0x13, {0x0C}, 1, 0},

{0x30, {0x00}, 1, 0},                 // Return to page 0

// Screen wake-up commands (delays required)
{0x11, {}, 0, 120},                   // Exit sleep mode, requires 120ms delay
{0x29, {}, 0, 20},                    // Turn on display, requires 20ms delay
```

5.3 Key Configuration Notes

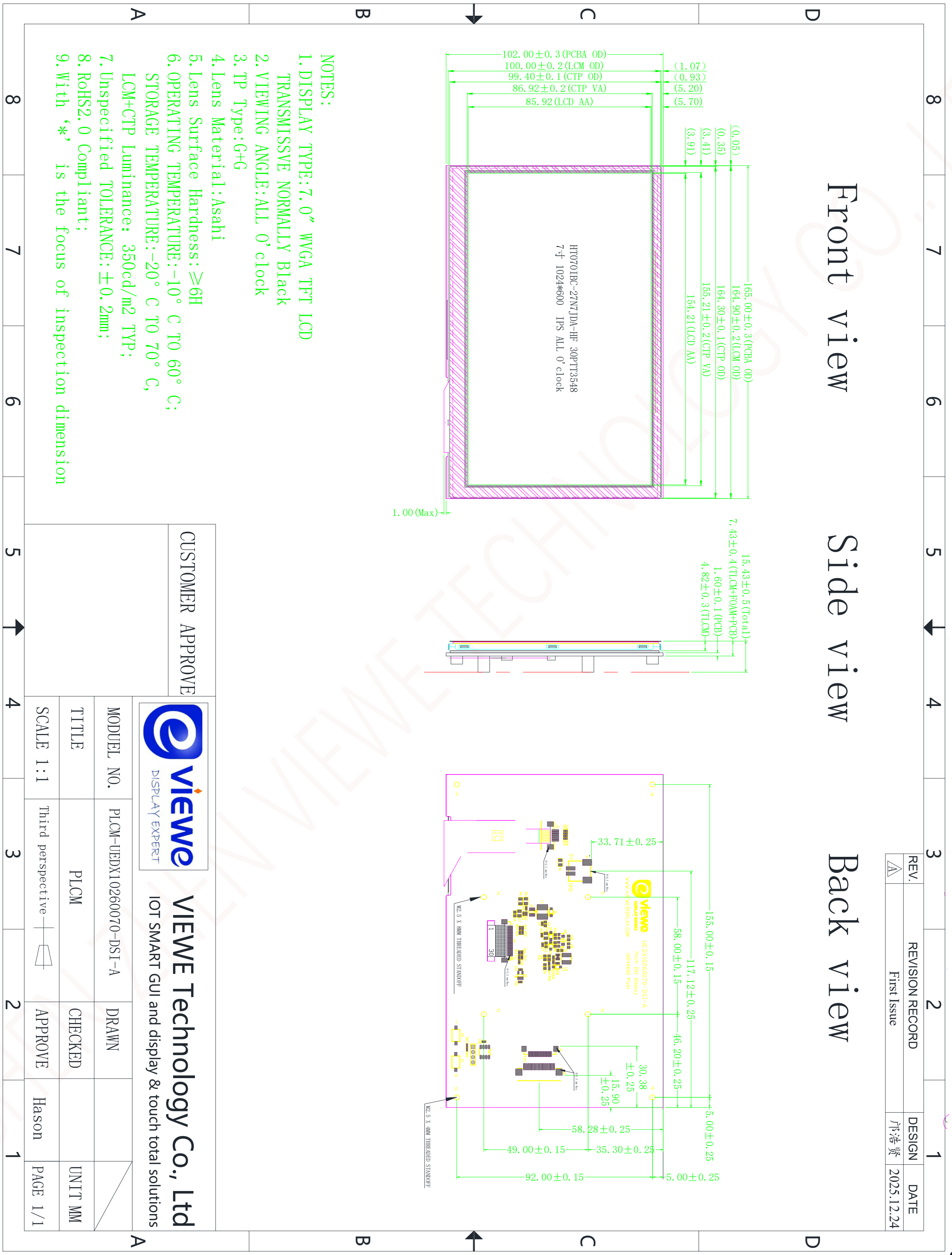
1. **MIPI Configuration:** ESP32P4 only supports 2 lanes mode, therefore register 0x0B must be set to 0x11
2. **Screen Resolution:** 1024×600 pixels
3. **Clock Frequency:** DPI clock 50MHz
4. **Gamma Correction:** Includes complete RGB gamma tables for accurate color reproduction
5. **Delay Requirements:** The last two commands (0x11 and 0x29) must be executed with specified delays

5.4 Usage Recommendations

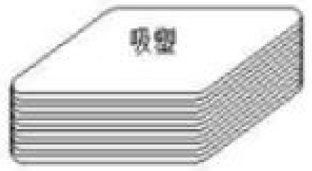
6. Send all commands in sequential order
7. Strictly adhere to delay requirements between commands
8. ESP32 platform must use 2 lanes configuration (0x0B = 0x11)
9. Timing parameters should be adjusted according to actual hardware connections
10. Ensure proper power sequencing before sending initialization commands

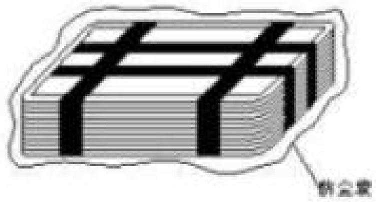
Note: The above structure takes the ESP32 platform as an example; please make corresponding adjustments for other platforms.

6. Mechanical Drawing



7. Package Drawing





第一步

将产品放入吸塑盘中，LCD AA 面朝上,注意防呆方向

第二步

每一层吸塑盘与相邻层，叠放时相错 180 度，最上层不放产品，总叠加层数参考

第三步

检查无误后用胶带固定吸塑盘，将捆好的吸塑盘放入无尘带中并封口；

First step

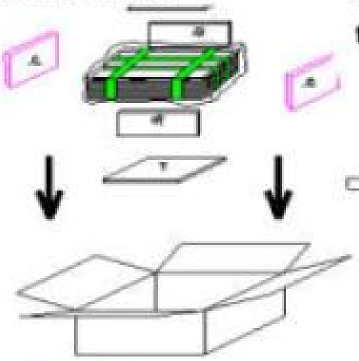
Putting products into the tray,
LCD A.A faces Upward,
(pay attention to the

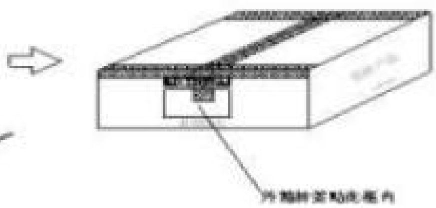
Second step

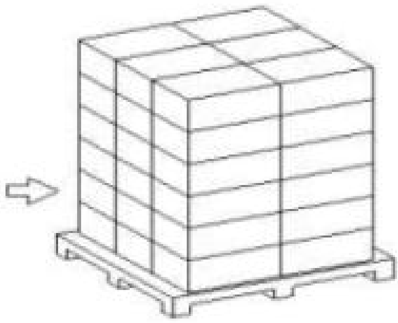
Neighbouring trays should be staggered 180° while stacking up.
on the top,there is an empty tray without product

Third step

Taping up the tray after inspection,and put them into a PE







第四步

外箱内侧底部和四周放上泡棉将包好的产品装入纸箱，合盖；

第五步

最后胶带封箱，贴外箱标签

第六步

将每箱整齐放在栈板上并包裹最高可堆叠 6 层)

Fourth step

Putting EPE foams and products with trays into the carton;
Close the carton box

fifth step

Sealing the carton with cellulose tape ;
Stick on a carton label,

sixth step

Placing the boxes together on a pallet (6 layers at most),